

Data Preparation

PRI 26/27 · Information Processing and Retrieval

M.EIC · Master in Informatics Engineering and Computation

M.IA · Master in Artificial Intelligence

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Today's Plan

- Focus on data preparation
 - Data preparation tasks
 - Overview of a typical pipeline
 - Example projects
- Review of the first milestone delivery
- Review the plan for the next practical class

Quick Review

Status

- Starting the 2nd week of classes.
- Lectures: **data collection** + **data preparation** >> data processing
- Practical: groups and topics >> **start working on data collection and preparation**
- Milestone 1 workshop **in 2 weeks.**

Project

- The development of an **information search system**, including work on
 - data collection and preparation,
 - information querying and retrieval,
 - and retrieval evaluation.
- Organized in **three milestones and workshops**:
 - M1: Data Preparation
 - M2: Information Retrieval
 - M3: Search System

Milestone #1: Data Preparation

- **Preparation and characterization** of the datasets selected for the project.
- This task is heavily dependent on the datasets, which may require some extraction actions such as crawling or scraping.
- Work on these tasks depends on the nature, volume, organization and accessibility of the selected datasets. As a result of this milestone, a well-documented and reproducible pipeline of data processing is expected.

Lab #2: Class Plan

- Data collection and processing.
- Tools exploration and evaluation (OpenRefine, Apache Tika, etc).
- Work on practical tutorials #1 (command line) and #2 (datasets).
- Continue work on project milestone 1.
 - Clear vision for the documents and information needs.
 - Obtaining datasets from the domain chosen for practical work.
 - Datasets storage.
 - Conduct exploratory data analysis.
 - Conceptual domain modeling.

Data Preparation

Data Preparation

- Real-world data is "messy".
- In practice, 50% to 80% of the time spent in data processing is spent in data preparation tasks.
- Data preparation, often called "data wrangling", captures activities like:
 - Understand what data is available;
 - Choose what data to use and at what level of detail;
 - Understand how to combine multiple sources of data;
 - Deciding how to distill the results to a size and shape that enables the follow-up steps.

Data Preparation

- After data properties are investigated and understood, a data preparation phase is generally needed to make the data suitable for the follow-up phases.
- Common data preparation tasks include:
 - Data **cleaning** — identify and fix data quality issues;
 - Data **transformation** — transform data to improve analysis or manipulation;
 - **Synthesis** of Data — create new attributes derived from existing data;
 - Data **integration** — combine data from different sources;
 - Data **reduction or selection** — eliminate data from the collection.

Data Transformation

- Data transformation operations can be performed to improve or facilitate data handling in subsequent stages.
- Transformation of data elements include:
 - **Normalization** of values to a comparable scale;
 - **Scaling** values to the same range (e.g. 0 to 1 for a follow-up method);
 - **Non-linear transformations** to deal with skewed distributions;
 - **Discretization or binning**, which transforms a numeric continuous value into ordered categorical values (e.g. having bins of the same size vs. having bins with the same number of items)
- Note that all operations over the data introduce layers of distortion and bias that are dependent on assumptions and interpretations. Documentation is key.

Discretization

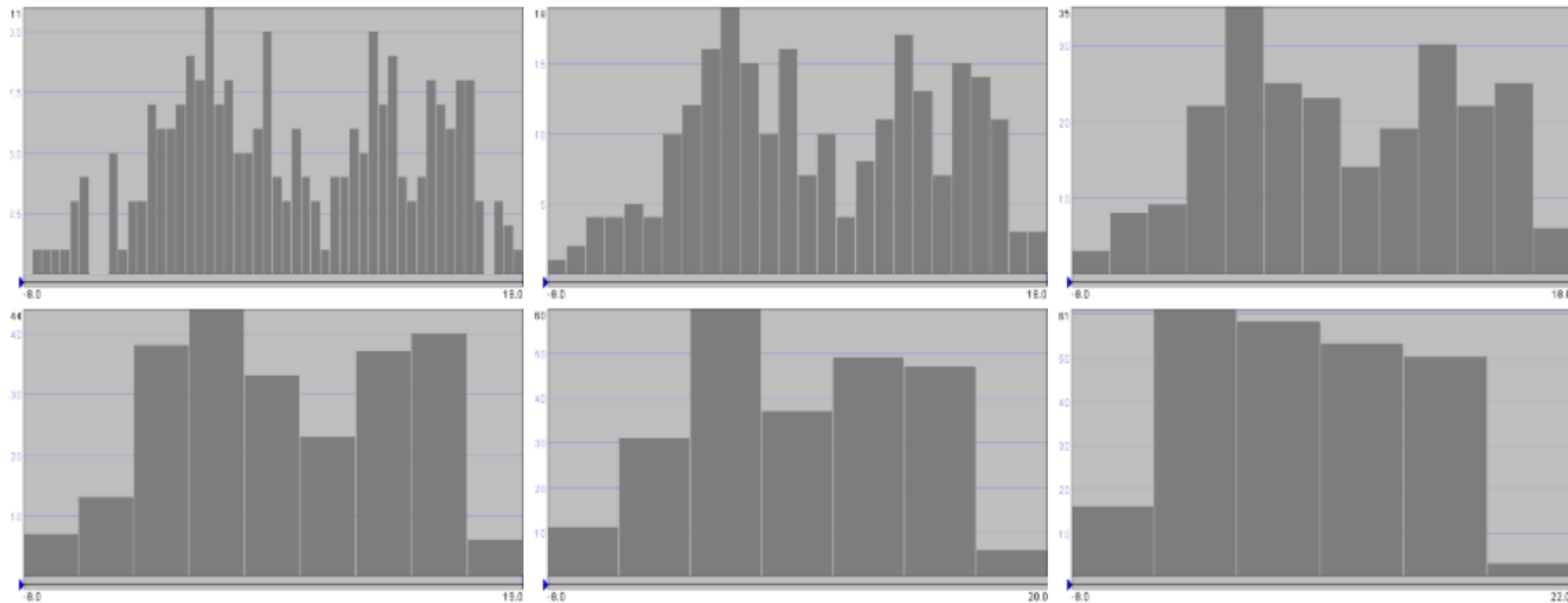


Fig. 3.13: Frequency histograms of the temperature data with the bin sizes equal to 0.5, 1, 2, 3, 4, and 5 degrees Celsius, correspondingly (left to right, top to bottom).

Synthesis of Data

- Combine existing attributes to produce new additional attributes that are more convenient to analyze or use as input in follow-up phases. Examples:
 - a new attribute representing the difference between two timestamps (e.g. duration);
 - the maximum value from a series of numerical attributes;
 - an integrated score that combines several attributes;
 - splitting an existing numerical series in two independent series (e.g., day and night);
 - most important keywords or topics extracted from a textual field;
 - etc.

Data Integration

- Combine data that originally exists in multiple sources.
- Key task in many projects, e.g. integrate data from multiple databases in an organization; enrich individual records with data from external sources.
- Complexity of data integration tasks differs significantly, from using join operations in a single database management system, to downloading, processing and linking external data sources.
- A central step of many data integration tasks is linking the corresponding records. A task that is easier if unique identifiers are available (rarely!), but of high complexity if based on other information (e.g., names, birth dates, addresses).

Data Reduction or Selection

- Data reduction or selection may be justified due to several reasons, namely:
 - data is not relevant;
 - data is outdated;
 - data volume exceeds the available capacity for processing it;
 - existing precision is excessive, while lower precision is sufficient.
- Techniques for performing data reduction or selection include:
 - data filtering
 - data sampling
 - data aggregation

Data Filtering

- Data filtering is used to remove data from the dataset, e.g. items with unsuitable values, data that is irrelevant for the scope of the project, outdated data items, data items that must be removed due to legal reasons, etc.
- Data filtering can also be used during development to test the planned approach using a manageable portion of the original collection.
- Data filtering operations are deterministic in nature, e.g. remove data from a given year, remove all references to a specific keyword, remove a specific data attribute, etc.

Data Sampling

- Data sampling is a non-deterministic process that takes a random subset of the data items of a requested size.
- When designing the sampling method it is important to ensure that the resulting sample is representative of the complete collection.
- Thus, it is important to analyze the distribution of data attributes before and after the sampling process.

Data Aggregation

- Data aggregation may be used to reduce excessive detail in data, decisions require:
 - choice of the grouping method, depending of domain-specific criteria;
 - selection of the aggregation operator (e.g. mean, median, min, max, percentile).

Visualization in Data Preparation

- Data preparation requires a good understanding of the data properties, thus data visualization is usually also used at this stage.
- The role of visualization at this stage is to visually present the result of the execution of different methods, e.g. outlier removal.
- Visualization in data preparation can be applied:
 - Before the application of computational methods to explore the properties of the data, e.g. choose an adequate computational method, detect disparities in data.
 - During the application of computational methods to inspect how data is being processed, e.g. observe intermediate data structures, track the data pipeline execution.
 - After the application of computational methods to evaluate the quality of the results, compare the results between different executions of the pipeline.

Tools for Data Preparation

Tools for Data Cleaning

- Existing tools combine visual and computational techniques for identifying and fixing data problems.
- A freely available open-source reference software is **OpenRefine** (formerly Google Refine)
 - <https://openrefine.org>
 - Focus on tabular data;
 - Supports data cleaning operations on numerical data, dates and time, and textual data;
 - Data exploration with descriptive statistics, facets, and histograms;
 - Data transformation with batch detection and replacement of inconsistent or missing values, time and date formatting, textual misspellings;
 - Integration with Wikidata for both obtaining and publishing data.
- Explore available tutorials in lab classes.

OpenRefine

OpenRefine

100 Most Populous Cities from Wikidata

Permalink

Open...

Export ▾

Help

Facet / Filter

Undo / Redo 0 / 0

Refresh

Reset All

Remove All

countryLabel

change

20 choices

Sort by: name count

Cluster

Bangladesh 1

Brazil 2

Chile 1

Democratic Republic of the Congo 1

India 8

Indonesia 1

Iran 1

Japan 2

Manchukuo 1

Nigeria 2

100 rows

Extensions: Wikidata ▾

Show as: rows records

Show: 5 10 25 50 rows

« first < previous 1 - 10 next > last »

All

cityLabel

population

countryLabel

☆	🗨	1.	Shanghai	23390000	People's Republic of China
☆	🗨	2.	Beijing	21710000	People's Republic of China
☆	🗨	3.	Lagos	21324000	Nigeria
☆	🗨	4.	Dhaka	16800000	Bangladesh
☆	🗨	5.	Mumbai	15414288	India
☆	🗨	6.	Istanbul	14657434	Turkey
☆	🗨	7.	Tokyo	13942856	Japan
☆	🗨	8.	Tianjin	13245000	People's Republic of China
☆	🗨	9.	Guangzhou	13080500	People's Republic of China
☆	🗨	10.	São Paulo	12106920	Brazil

55 records

Schema *

Issues

Preview

Extensions: Wikidata ▾

The Wikidata schema below specifies how your tabular data will be transformed into Wikidata edits. You can drag and drop the column names below in most input boxes: for each row, edits will be generated with the values in these columns.

Save schema

Discard changes

Author

Title

Publication year

Literary genre

Country

Reference

Title

remove

Terms

no labels, descriptions or aliases added

+ add term

Statements

author

Author

remove

+ add qualifier

1 references

copy

reference URL

Referen...

remove

remove

+ add

+ add reference

+ add value

publication date

Publication y...

remove

+ add qualifier

1 references

copy

reference URL

Referen...

remove

remove

+ add

+ add reference

+ add value

genre

Literary ge...

remove

+ add qualifier

1 references

copy

reference URL

Referen...

remove

remove

+ add

+ add reference

+ add value

+ add statement

Tools for Data Preparation

- **Parse and extract text and metadata from files**
 - Apache Tika (e.g. PDF, PPT, XLS), <https://tika.apache.org>
 - BeautifulSoup (HTML), <https://www.crummy.com/software/BeautifulSoup/>
- NLP toolkits provide **natural text processing tools**
 - spaCy, <http://spacy.io>
 - NLTK, <http://nltk.org>
 - Apache OpenNLP, <http://opennlp.apache.org>
- **Data processing**
 - R, <http://www.r-project.org>
 - Pandas, <https://pandas.python.org>

Cloud Computing

- Cloud-based services are not a focus of this course but are a relevant piece in modern data processing and analytics pipelines.
- Managed services make building and deploying data pipelines more accessible, due to the elastic properties of the service, and the outsourcing of the setup and configuration part.
- Models based on pay-per-use, enable setting up and discarding data processing pipelines as needed.
- Cloud solutions are a good option to ad-hoc, highly diverse needs.
- However, the **risk of lock-in** is central and must be taken into consideration.

Data Pipelines

Data Pipelines

- Data pipelines are sets of processes that move and transform data from various sources to various destinations where new value can be derived.
- Complexity of data pipelines varies widely, depending on size, state, structure of the data sources as well as the needs of the project.
 - **Simple example:** obtain data from a REST API and load it into a relational database;
 - **Complex example:**
 - obtain data from multiple web pages at regular intervals;
 - parse HTML and extract main keyword from each page;
 - automatically identify main trending topics using a previously trained machine learning model;
 - update model with new data (keep an archive of previous models);
 - integrate topics and web pages in a cloud-based storage linked to a live dashboard.

Characteristics of a Data Pipeline

- A data pipeline is a software system; thus, general software best practices apply. Those highlighted here are particularly relevant in the context of data pipelines.
 - **Reliable** — work as expected even in face of adversity (software, hardware, or human faults).
 - **Scalable** — cope with increased load (in volume, traffic, complexity) in a manageable way (e.g., adding resources, distributing load).
 - **Maintainable** — evolve through time and teams (reduce complexity, make changes easier).
- Data collection and preparation is usually an ad-hoc and exploratory process, easily leading to a dispersion in threads and activities. Adoption of pipeline management systems and a complete and detailed documentation is key.
- Treating data processing pipelines as software makes them more maintainable, versionable, testable and collaborative.

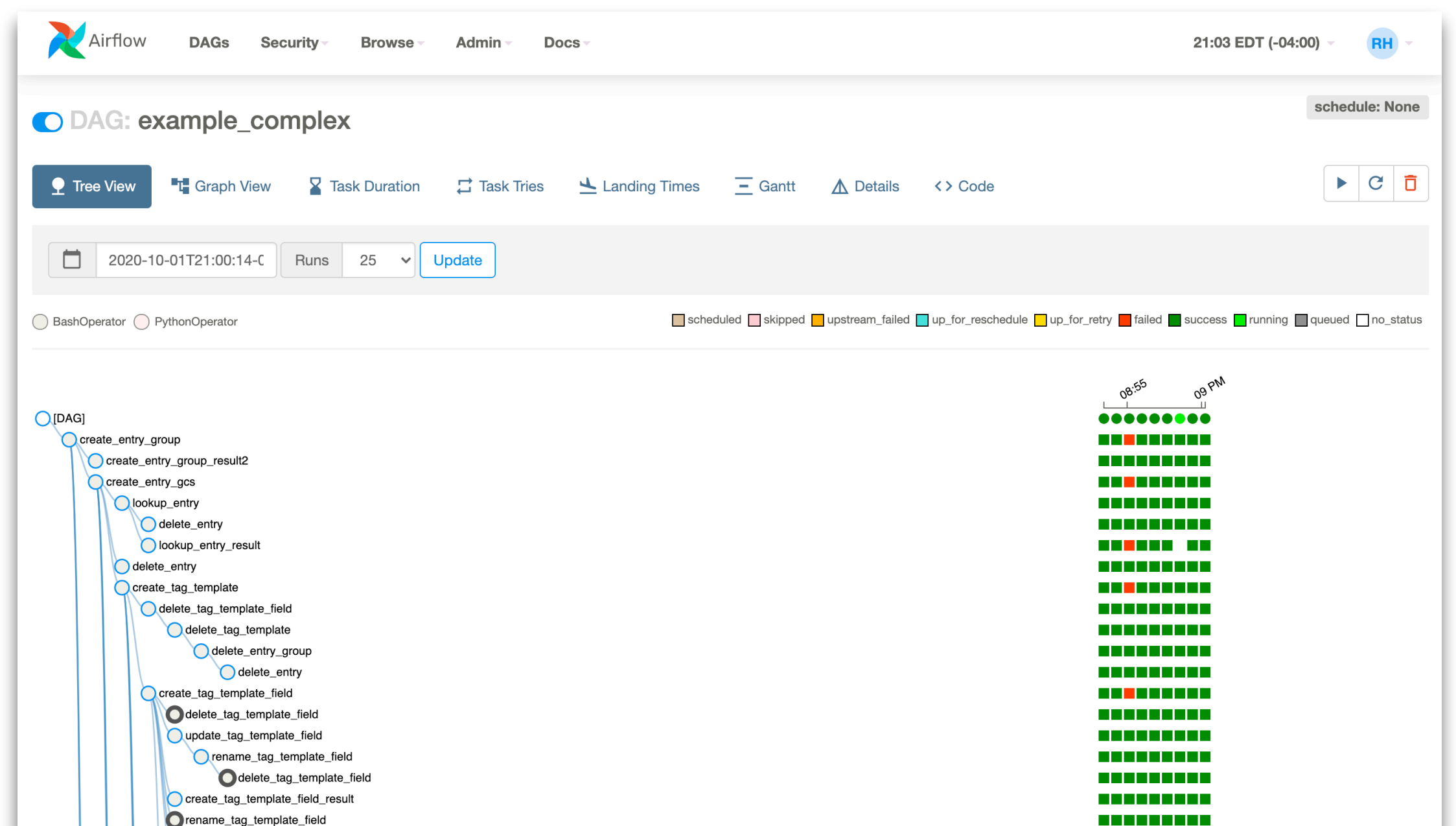
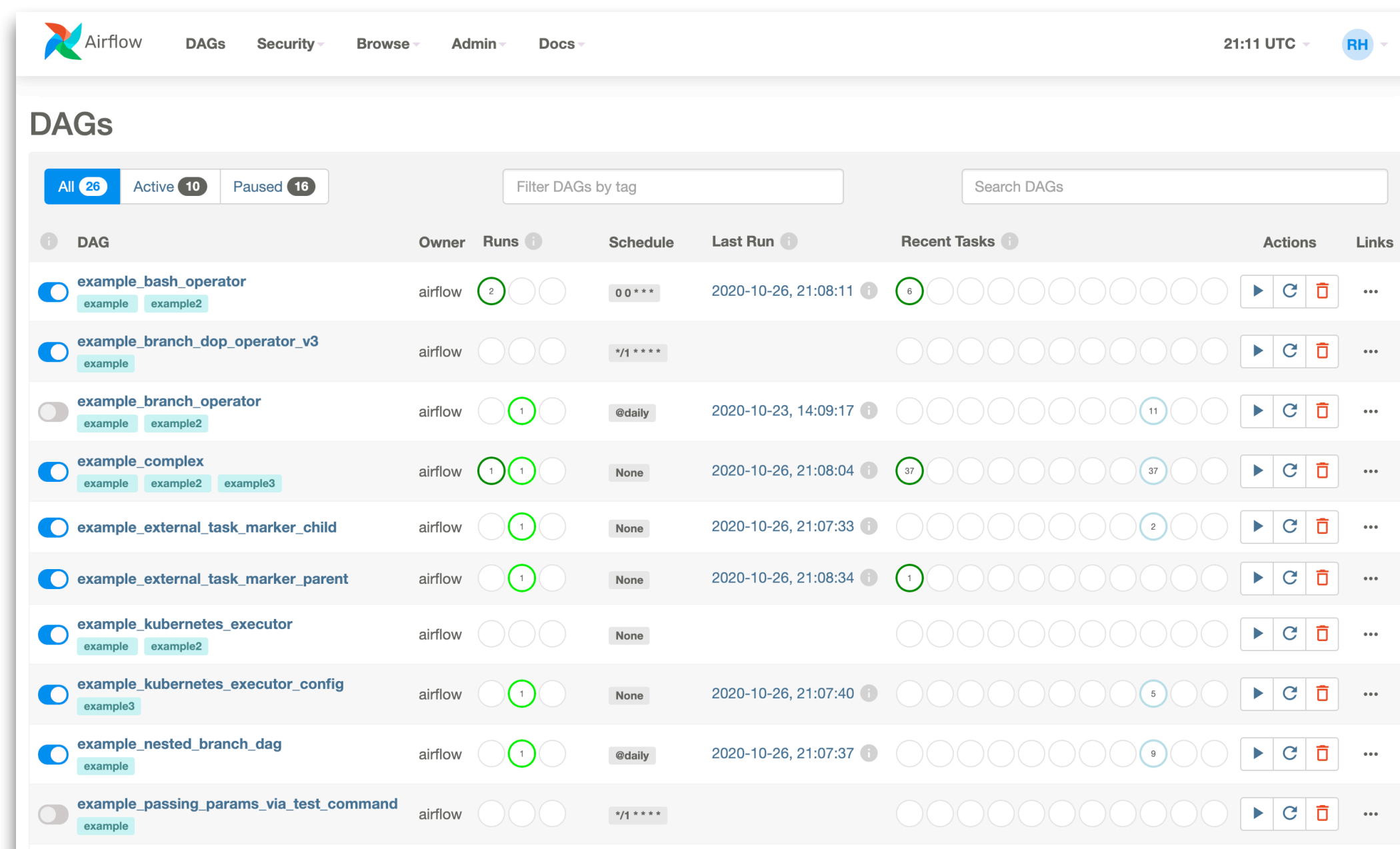
Tools for Data Pipelines

Makefiles

- Makefiles are used to automate software build processes, by defining targets and rules to execute. The underlying abstraction is of a dependency graph, where tasks depend on the execution of other tasks.
- Make language agnostic, and implementations exist for most operating systems.
- Can be used to document and setup data pipelines.
- Not mandatory for the PRI project, but a good practice.
- A tutorial is available to explore in lab classes.

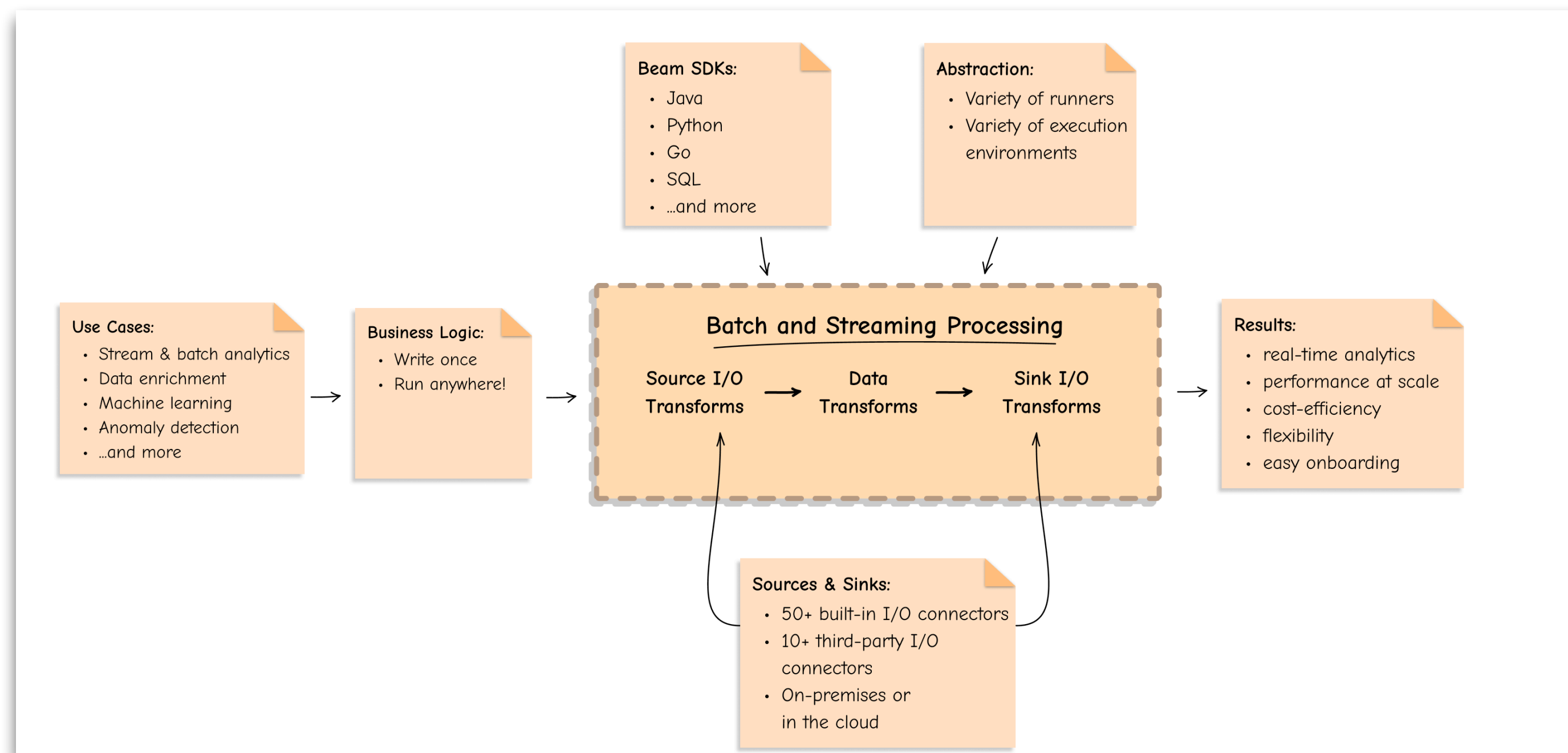
Apache Airflow

- An open-source platform, to "programmatically author, schedule and monitor workflows" [airflow.apache.org]. Supports Python and Bash scripts.



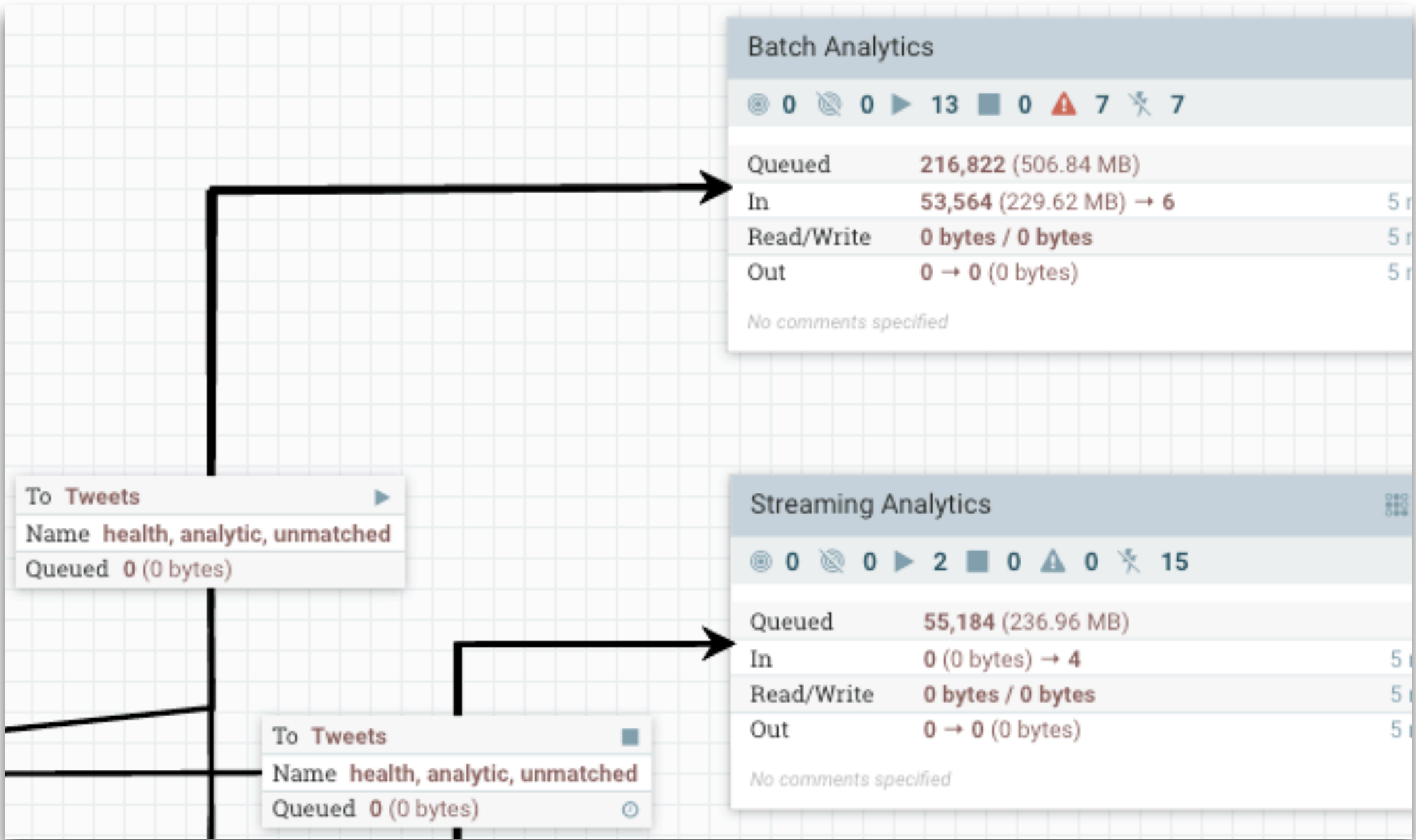
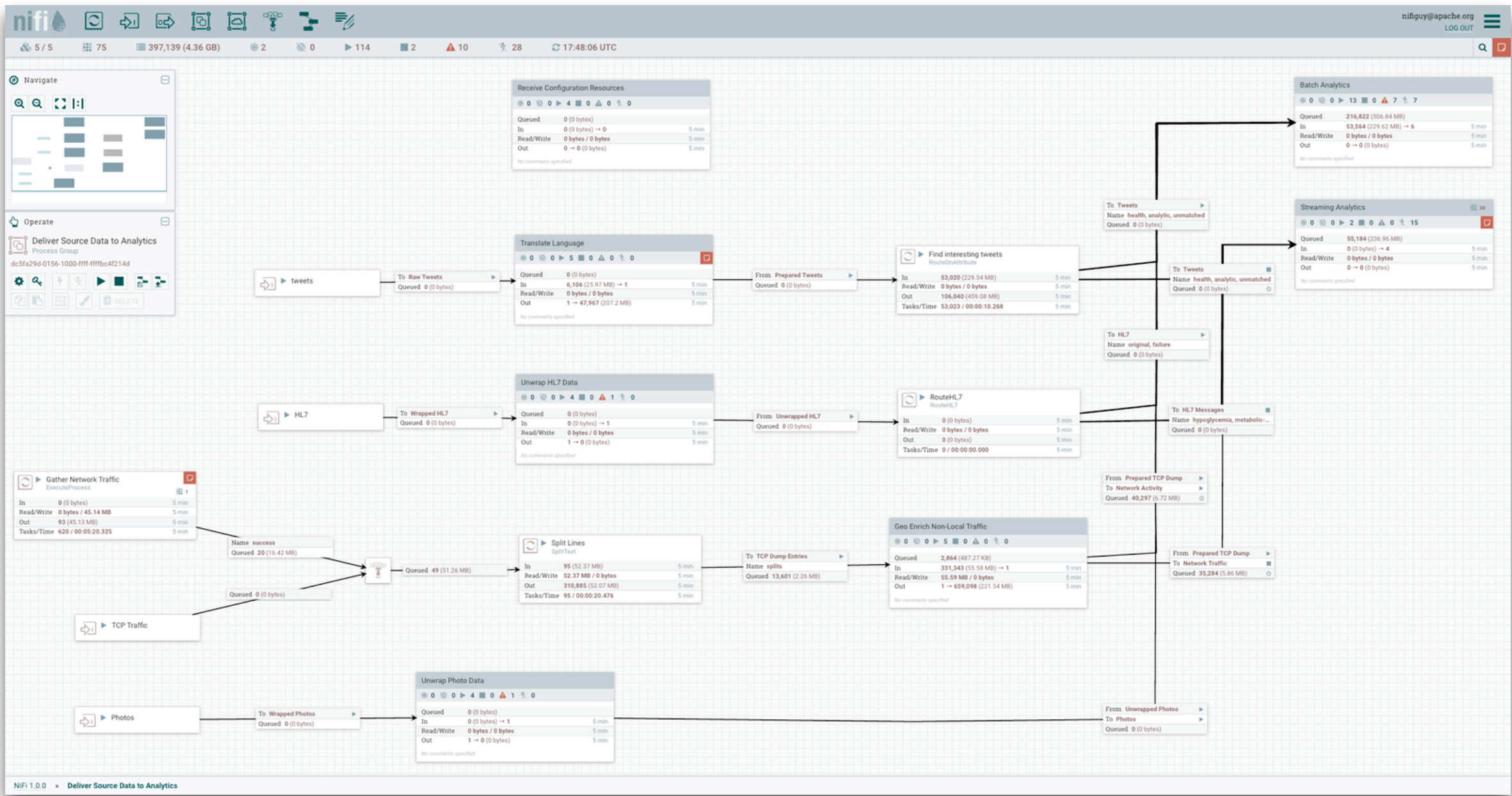
Apache Beam

- "Apache Beam is an open-source, unified programming model for batch and streaming data processing pipelines that simplifies large-scale data processing dynamics." [beam.apache.org]



Apache NiFi

- "Apache NiFi is designed to automate the flow of data between software systems, leveraging the ETL pattern (extract, transform, load)." [nifi.apache.org]



Data Streams Processing

- **Apache Kafka**, kafka.apache.org
 - "An open-source distributed event store and stream-processing platform, providing a unified, high-throughput, low-latency platform for handling real-time data feeds."
- **Apache Flink**, flink.apache.org
 - "A framework and distributed processing engine for stateful computations over unbounded and bounded data streams, designed to run in all common cluster environments, perform computations at in-memory speed and at any scale."

Documentation of Data Pipelines

Data Documentation

- Documentation is central in any engineering process.
- It distinguishes between ad-hoc processes and repeatable, inspectable, shareable processes.
- It is essential to document:
 - Data selection and sampling criteria, sources, filtering, etc.
 - Data formats across the pipeline (input, output, as well as intermediary files);
 - Platform and software versions (OS, tools);
 - Operations performed (extract, sample, link, etc);
- For understanding underlying assumptions (bias) in data, documentation is a key element.

Data Flow Diagrams (DFD)

- Data-flow diagrams can be used to represent the flow of data from external entities into the system, show how data moves from one process to another, and data's logical storage.
- The notation includes four symbols:
 - **Squares** represent external entities, i.e. sources or destinations of data
 - **Rounded rectangles** represent processes, which takes data as input, perform operations over it, and then output it.
 - **Arrows** represent data flows, i.e. how data moves around.
 - **Open-ended rectangles** represent data stores, e.g. databases, files.

Bibliography and Further Reading

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